

Gaussian Confounding and Sharp Recovery of Uniform Convolution Factors

Shifted power sums, minimax estimation, and detection
above a Fabius–Rvachev background

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Abstract

Let a known bounded random variable be convolved with at most m centered uniform factors and a Gaussian whose variance may be unknown. We determine the exact global minimax rates for recovering the ordered uniform half-lengths from independent observations. The rate is $n^{-1/(4m)}$ when the Gaussian variance is known and $n^{-1/(4m+4)}$ when it is unknown. In the latter model, the optimal rate for the Gaussian variance itself is $n^{-1/(2m+2)}$. The constants may depend on the fixed capacity m , the background, and a compact positive variance interval.

The algebraic engine is a sharp shifted-power-sum inequality: the moments of orders $r+1, \dots, r+m$ determine m ordered nonnegative nodes with optimal inverse Hölder exponent $1/(m+r)$. We prove this directly using sign changes of an integer-valued counting function, and construct exact moment-preserving curves attaining the exponent. Gaussian compensation converts the missing first power sum into a genuine loss of statistical information. A weighted likelihood expansion supplies matching Hellinger and chi-square asymptotics, closing the gap between algebraic moment cancellation and statistical lower bounds.

Detection has a different scale: the presence of any nonzero uniform factor has critical separation $n^{-1/8}$ with unknown Gaussian variance, independent of m , versus $n^{-1/4}$ with known variance. The Rvachev up law is an explicit admissible background. The article includes proofs, a feasible moment estimator, exact finite algebraic checks, a source comparison, and a research agenda. These are conventional mathematical proofs, not a Lean verification or a claim of worldwide publication priority.

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1 Research target and relation to ProveIt

1.1 The extension addressed here

A convolution factorization can be identifiable while its recovery from noisy data is extremely unstable. The ProveIt repository develops this question for Fabius–Rvachev laws and more general uniform convolutions. Its recent report *Sharp Stability Strata for Finite Fabius–Rvachev Deconvolution* [5] studies at most m unknown uniform factors over a known smooth compactly supported background. It gives deterministic total-variation inverse exponents, explicitly distinguishes pairwise and fixed-base stability, and explicitly does not establish a statistical minimax lower bound. The repository index records that report as an unreviewed research intake rather than a Lean-verified theorem [6].

We change the statistical experiment in an essential way: there is a *positive Gaussian convolution*, and its variance can be an unknown nuisance parameter. Gaussian noise contributes only to the second cumulant. Consequently, it can conceal the variance of small uniform factors while leaving their higher cumulants visible. How much statistical information does this remove? Can one recover the noise variance and all the factors at matching upper and lower rates? Is detecting a factor as hard as recovering the entire factorization? These are the concrete questions answered below.

Our conclusions are not asserted for the unsmoothed experiment of the preceding report. Compact-support boundaries create additional likelihood issues there. Positive Gaussian smoothing is a hypothesis, not a disposable technical convenience. Conversely, we require only that the known background be bounded; it need not have a density or be symmetric.

1.2 Classical ingredients and the contribution boundary

Cumulants, Newton identities, Vandermonde matrices, and the geometry of singular moment inversion are classical. Prony-map singularities provide closely related algebraic context [2]. Moment fitting and moment-matching statistical lower bounds have an extensive literature for finite mixtures; Heinrich–Kahn and Wu–Yang are relevant examples [3, 4]. Those mixture models have unknown component locations and usually unknown mixing weights. Our model instead has a *sum of independent uniforms*, with fixed unit multiplicities in its power sums. Mixture theorems are not being applied as though they directly proved the rates below.

The results developed here are a sharp consecutive shifted-moment inequality, an explicit variance-confounding moment flow, a weighted likelihood expansion, and matching estimation and testing bounds for this convolution model. All proofs needed for those conclusions are supplied. The source comparison is focused, not an exhaustive worldwide novelty audit; no claim of publication priority is made. None of the new proofs has been independently refereed or checked in Lean or Rocq.

The repository snapshot used for the principal comparison is

c130dba623c420551d90db41dae7b3d9cff72dfd.

The accompanying `SOURCE_AUDIT.md` records the exact paths and scope. Search-index hits at other commits were used for discovery, not silently identified with this snapshot. No repository files were modified.

2 Model and main statistical results

Fix an integer $m \geq 1$ and a known random variable Z with $|Z| \leq L$ almost surely. Let $U_1, \dots, U_m$ be independent uniforms on $[-1, 1]$, and let G be a standard normal, all independent of Z . Write

$$\mathcal{A}_m = \{a \in [0, 1]^m : a_1 \leq \dots \leq a_m\}, \quad X_{a,v} = Z + \sum_{i=1}^m a_i U_i + \sqrt{v} G, \quad P_{a,v} = \mathcal{L}(X_{a,v}). \quad (2.1)$$

Zero entries pad the list to capacity m ; they do not represent observable factors. The variance parameter is either fixed at a known $v_0 > 0$, or ranges in a fixed interval

$$0 < \underline{v} < \bar{v} < \infty. \quad (2.2)$$

We observe n independent copies of $X_{a,v}$. Ordered maximum error is the loss for a , and absolute error is the loss for v . Thus, in the unknown variance model, define

$$\mathcal{R}_n^a = \inf_{\hat{a}} \sup_{a \in \mathcal{A}_m, v \in [\underline{v}, \bar{v}]} \mathbb{E}_{a,v} \|\hat{a} - a\|_\infty, \quad (2.3)$$

$$\mathcal{R}_n^v = \inf_{\hat{v}} \sup_{a \in \mathcal{A}_m, v \in [\underline{v}, \bar{v}]} \mathbb{E}_{a,v} |\hat{v} - v|. \quad (2.4)$$

Estimators may be measurable functions of the entire sample. Let $\mathcal{R}_n^{a,\text{known}}$ denote the analogous scale risk with $v = v_0$ fixed. All asymptotic constants below can depend on fixed m, L, Z and the variance bounds (or v_0), but not on n or the varying parameters.

Theorem 2.1 (Sharp global estimation rates). *There are positive constants c, C and n_0 such that, for $n \geq n_0$,*

$$cn^{-1/(4m+4)} \leq \mathcal{R}_n^a \leq Cn^{-1/(4m+4)}, \quad (2.5)$$

$$cn^{-1/(2m+2)} \leq \mathcal{R}_n^v \leq Cn^{-1/(2m+2)}, \quad (2.6)$$

$$cn^{-1/(4m)} \leq \mathcal{R}_n^{a,\text{known}} \leq Cn^{-1/(4m)}. \quad (2.7)$$

A single moment-fitting procedure attains both upper bounds in the unknown variance model.

Capacity	Known-variance scales	Unknown-variance scales	Unknown variance
$m = 1$	$n^{-1/4}$	$n^{-1/8}$	$n^{-1/4}$
$m = 2$	$n^{-1/8}$	$n^{-1/12}$	$n^{-1/6}$
$m = 3$	$n^{-1/12}$	$n^{-1/16}$	$n^{-1/8}$
General m	$n^{-1/(4m)}$	$n^{-1/(4m+4)}$	$n^{-1/(2m+2)}$

For detection, compare the null $a = 0$ with alternatives $\|a\|_\infty \geq h$. The variance is allowed to vary under both hypotheses when it is unknown. Write $\mathcal{T}_n(h)$ for the infimum, over tests, of the sum of the maximal type-I and type-II errors.

Theorem 2.2 (Detection boundary). *In the unknown variance model, there is a test with $\mathcal{T}_n(h) \leq C/(nh^8)$. If $h_n \rightarrow 0$ and $nh_n^8 \rightarrow 0$, then $\mathcal{T}_n(h_n) \geq 1 - o(1)$. With known variance, the corresponding statements hold with h^4 in place of h^8 .*

Thus consistent detection is possible when $nh^8 \rightarrow \infty$ and impossible when $nh^8 \rightarrow 0$ in the unknown-variance experiment. The critical order is $n^{-1/8}$, independently of m . For $m > 1$, recovering every half-length to its worst-case optimal precision is strictly harder than detecting the presence of at least one factor. Proofs appear in Sections 8, 9, and 10.

3 Cumulants and the missing first power sum

For a vector $x = (x_1, \dots, x_m)$, let $p_k(x) = \sum_i x_i^k$ and put $x_i = a_i^2$. All random variables in our model have moment generating functions in a neighborhood of the origin. Cumulants are defined by

$$\log \mathbb{E} e^{tX} = \sum_{j \geq 1} \kappa_j(X) \frac{t^j}{j!}. \quad (3.1)$$

For U uniform on $[-1, 1]$, $\mathbb{E}e^{tU} = \sinh(t)/t$. Its even cumulants are

$$c_k := \kappa_{2k}(U) = \frac{2^{2k} B_{2k}}{2k} = (-1)^{k+1} \frac{(2k)! \zeta(2k)}{k \pi^{2k}}, \quad k \geq 1, \quad (3.2)$$

where the Bernoulli convention is $B_2 = 1/6$. The formula follows by integrating the Taylor series of $\coth t - 1/t$, or by expanding the product for $\sinh(t)/t$. In particular,

$$c_1 = \frac{1}{3}, \quad c_2 = -\frac{2}{15}, \quad c_3 = \frac{16}{63}; \quad c_k \neq 0 \text{ for every } k. \quad (3.3)$$

Independence and scaling give the exact identities

$$\kappa_2(X_{a,v}) - \kappa_2(Z) = v + \frac{1}{3}p_1(x), \quad (3.4)$$

$$\kappa_{2k}(X_{a,v}) - \kappa_{2k}(Z) = c_k p_k(x), \quad k \geq 2. \quad (3.5)$$

All odd cumulants of the added noise are zero. If v is known, p_1 is observed through the second cumulant. If v is unknown, only the combination $v + p_1/3$ is observed there. The available uncontaminated consecutive power sums begin at p_2 , not p_1 .

3.1 The Fabius–Rvachev specialization

Arias de Reyna discusses the smooth compactly supported up function and its infinite product representation [1]. For our normalization it suffices to define the background probabilistically:

$$Z_{\text{up}} = \sum_{j=1}^{\infty} 2^{-j} U_j. \quad (3.6)$$

The series converges absolutely for every outcome and has support in $[-1, 1]$. Its characteristic function is

$$\mathbb{E}e^{itZ_{\text{up}}} = \prod_{j=1}^{\infty} \text{sinc}(2^{-j}t), \quad \text{sinc } t = \frac{\sin t}{t}, \quad \text{sinc } 0 = 1. \quad (3.7)$$

Bounded convergence gives the product, and locally convergent logarithms give

$$\kappa_{2k}(Z_{\text{up}}) = \frac{c_k}{2^{2k} - 1}. \quad (3.8)$$

In particular, its variance is $1/9$ and its fourth cumulant is $-2/225$. These formulas make the required background cumulants explicit. Our proofs use boundedness, not any unproved regularity claim about the up density.

4 A sharp inverse theorem for shifted power sums

Let $\mathcal{X}_m = \{x \in [0, 1]^m : x_1 \leq \dots \leq x_m\}$. For an integer $r \geq 0$ define

$$E_r(x, y) = \max_{r+1 \leq k \leq r+m} |p_k(x) - p_k(y)|. \quad (4.1)$$

Theorem 4.1 (Consecutive shifted power sums). *For all $x, y \in \mathcal{X}_m$ and all integers $r \geq 0$,*

$$\|x - y\|_{\infty} \leq 8E_r(x, y)^{1/(m+r)}. \quad (4.2)$$

The exponent $1/(m+r)$ cannot be replaced by any larger exponent in a uniform bound on $\mathcal{X}_m$. Consequently, the map $x \mapsto (p_{r+1}(x), \dots, p_{r+m}(x))$ is injective.

Proof of the upper bound. Set

$$C(t) = \#\{i : x_i \leq t\} - \#\{i : y_i \leq t\}, \quad 0 < t < 1.$$

This is an integer-valued step function. Its nonzero sign changes at most $m - 1$ times. To see this even when atoms coincide, split its jumps into unit up-steps and down-steps. There are m of each, and the resulting walk starts and ends at zero. Each excursion from zero consumes at least one up-step and one down-step, so there are at most m excursions. Consecutive nonzero sign blocks of C can only decrease this count. Intermediate steps at a common atom occupy no interval and do not create an extra sign block of C .

Choose points $z_1, \dots, z_s \in [0, 1]$ between consecutive nonzero sign blocks of opposite sign, with $s \leq m - 1$. There is a sign choice such that

$$Q(t) = \pm \prod_{j=1}^s (t - z_j), \quad Q(t)C(t) \geq 0$$

almost everywhere. If there are no sign changes, Q is the appropriate constant 1 or -1 . The sum of the absolute values of the coefficients of Q is at most 2^s .

Put $h = \|x - y\|_\infty$. There is nothing to prove for $h = 0$. By interchanging x, y if necessary, some i satisfies $y_i - x_i = h$. For $x_i < t < y_i$, $C(t) \geq 1$. No z_j lies in this open interval. On its middle half, which has length $h/2$, we therefore have $t \geq h/4$ and $|t - z_j| \geq h/4$ for every j . It follows that

$$\int_0^1 t^r Q(t)C(t) dt \geq \frac{h}{2} \left(\frac{h}{4}\right)^{r+s}. \quad (4.3)$$

For $k \geq 1$, direct integration of the counting functions gives

$$p_k(x) - p_k(y) = -k \int_0^1 t^{k-1} C(t) dt. \quad (4.4)$$

Writing Q in monomials and using $s \leq m - 1$ yields

$$\left| \int_0^1 t^r Q(t)C(t) dt \right| \leq \frac{2^s}{r+1} E_r(x, y). \quad (4.5)$$

Thus $h^{r+s+1} \leq 2^{2r+3s+1} E_r/(r+1)$. Since $0 < h \leq 1$ and $s \leq m - 1$, we obtain

$$h^{m+r} \leq \frac{2^{2r+3m-2}}{r+1} E_r(x, y) \leq 8^{m+r} E_r(x, y),$$

which proves (4.2). $\square$

The proof uses nonnegativity of the nodes and their fixed unit weights. An arbitrary signed or unequal-weight moment problem is not covered by this argument. In particular, changing the model can change its critical exponent.

Corollary 4.2 (Half-length inversion). *For $a, b \in \mathcal{A}_m$ and $x_i = a_i^2, y_i = b_i^2$,*

$$\|a - b\|_\infty \leq \sqrt{8} E_r(x, y)^{1/(2m+2r)}. \quad (4.6)$$

This exponent is also sharp.

Proof. Use $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$. Sharpness follows from the scaled positive-node pairs constructed in the next section. $\square$

5 Exact moment fibres and optimality of the exponents

5.1 A general construction for every shift

Lemma 5.1 (A moment-preserving tangent). *Let $0 < u_1 < \dots < u_m < 1$ and $P(z) = \prod_i (z - u_i)$. For $r \geq 0$ set*

$$b_i = \frac{1}{u_i^r P'(u_i)}. \quad (5.1)$$

The differentials of $p_{r+1}, \dots, p_{r+m-1}$ vanish on b , while $Dp_{r+m}(u)b = r + m$. These first $m - 1$ differentials are linearly independent.

Proof. Lagrange interpolation of z^j , followed by comparison of the coefficient of z^{m-1} , gives

$$\sum_{i=1}^m \frac{u_i^j}{P'(u_i)} = \begin{cases} 0, & 0 \leq j < m-1, \\ 1, & j = m-1. \end{cases} \quad (5.2)$$

Now $Dp_k(u)b = k \sum_i u_i^{k-1-r} / P'(u_i)$, proving the asserted values. For rank, factor the nonzero u_i^r from each column of the derivative matrix and the nonzero row constants. What remains is a rectangular Vandermonde matrix of rank $m - 1$. The statements include $m = 1$, with no constraints and an empty rank assertion. $\square$

The implicit function theorem gives a smooth one-dimensional local curve through u , with tangent b , on which $p_{r+1}, \dots, p_{r+m-1}$ are exactly constant. Along this curve p_{r+m} has nonzero derivative. Choose another point w sufficiently close to u so that all coordinates remain distinct and in $(0, 1)$, and $p_{r+m}(u) \neq p_{r+m}(w)$. Then, for $0 < \lambda \leq 1$,

$$\|\lambda u - \lambda w\|_\infty = \lambda \|u - w\|_\infty, \quad (5.3)$$

$$E_r(\lambda u, \lambda w) = \lambda^{m+r} |p_{m+r}(u) - p_{m+r}(w)|. \quad (5.4)$$

A bound with exponent $\alpha > 1/(m+r)$ would force a positive constant times λ to be at most a constant times $\lambda^{(m+r)\alpha}$ as $\lambda \downarrow 0$, which is impossible. This proves the sharpness clause of Theorem 4.1. Applying the same pair to half-lengths $\sqrt{\lambda}\sqrt{u_i}$ and $\sqrt{\lambda}\sqrt{w_i}$ proves Corollary 4.2's sharpness.

5.2 An explicit flow that changes only the first visible deficit

The general construction suffices for sharpness, but statistical variance recovery needs control of p_1 as well. For shift $r = 1$, the following coefficient flow provides it without an unspecified choice of tangent.

Proposition 5.2 (First-power-sum flow). *Let $0 < u_1 < \dots < u_m < 1$, and let $e_k(0)$ be their elementary symmetric functions, with $e_0(0) = 1$. Define*

$$e_k(s) = \sum_{j=0}^k \frac{s^j}{j!} e_{k-j}(0), \quad P_s(z) = \sum_{k=0}^m (-1)^k e_k(s) z^{m-k}. \quad (5.5)$$

For all sufficiently small real s , the roots $u_i(s)$ of P_s remain simple, ordered, and in $(0, 1)$. Their power sums satisfy

$$p_1(u(s)) = p_1(u(0)) + s, \quad (5.6)$$

$$p_k(u(s)) = p_k(u(0)), \quad 2 \leq k \leq m, \quad (5.7)$$

$$\frac{d}{ds} p_{m+1}(u(s)) = (-1)^{m+1} (m+1) e_m(s) \neq 0. \quad (5.8)$$

Proof. Simple real roots continue smoothly by the implicit function theorem; choose s small enough that disjoint root neighborhoods lie in $(0, 1)$. Set $E_s(t) = \sum_{k=0}^m e_k(s)t^k$. Then

$$E_s(t) = e^{st}E_0(t) \pmod{t^{m+1}}, \quad \partial_s E_s(t) = tE_s(t) - e_m(s)t^{m+1}. \quad (5.9)$$

As a formal power series at zero,

$$\log E_s(t) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} p_k(u(s))t^k. \quad (5.10)$$

Taking logarithms in the first identity gives the first two assertions. Dividing the second identity by $E_s(t)$ and taking the coefficient of t^{m+1} gives the third. Since all roots are positive, $e_m(s) > 0$. $\square$

The flow is exact, not just a cancellation of first derivatives. Choosing a fixed small nonzero s gives two configurations u, w with

$$p_2(u) = p_2(w), \dots, p_m(u) = p_m(w), \quad p_1(u) \neq p_1(w), \quad p_{m+1}(u) \neq p_{m+1}(w). \quad (5.11)$$

These are the hard pairs used below. They explain concretely how a change in the unknown Gaussian variance can hide a change in the first power sum while many additional cumulants also agree.

6 Identifiability and deterministic Hellinger stability

For densities f, g , use the conventions

$$H^2(f, g) = \int (\sqrt{f} - \sqrt{g})^2 dx, \quad \chi^2(f, g) = \int \frac{(f - g)^2}{g} dx, \quad d_{TV}(f, g) = \frac{1}{2} \int |f - g| dx. \quad (6.1)$$

Here $0 \leq H^2 \leq 2$, and $d_{TV} \leq H$.

Proposition 6.1 (Finite-cumulant identifiability). *In the unknown variance model, cumulants of orders $2, 4, 6, \dots, 2m + 2$ determine (a, v) uniquely. With known variance, orders $2, 4, \dots, 2m$ determine a uniquely.*

Proof. Use (3.5) to obtain $p_2, \dots, p_{m+1}$ when v is unknown. Theorem 4.1 with $r = 1$ determines the ordered x ; then $a_i = \sqrt{x_i}$ and (3.4) determines v . For known variance use $r = 0$ and $p_1, \dots, p_m$ instead. $\square$

Theorem 6.2 (Global Hellinger inverse moduli). *On the parameter sets of Section 2,*

$$\|a - b\|_\infty \leq CH(P_{a,v}, P_{b,w})^{1/(2m+2)}, \quad (6.2)$$

$$|v - w| \leq CH(P_{a,v}, P_{b,w})^{1/(m+1)}. \quad (6.3)$$

If $v = w = v_0$ is known, the stronger scale estimate is

$$\|a - b\|_\infty \leq CH(P_{a,v_0}, P_{b,v_0})^{1/(2m)}. \quad (6.4)$$

All three exponents are optimal for uniform bounds on these parameter sets.

Proof of the upper estimates. All absolute moments of any fixed order are uniformly bounded in the model. For raw moments of order j , Cauchy–Schwarz gives

$$\left| \int x^j (f - g) dx \right| \leq H(f, g) \left(\int |x|^{2j} (\sqrt{f} + \sqrt{g})^2 dx \right)^{1/2} \quad (6.5)$$

$$\leq H(f, g) (2\mathbb{E}_f |X|^{2j} + 2\mathbb{E}_g |X|^{2j})^{1/2} \leq C_j H(f, g). \quad (6.6)$$

Cumulants are polynomials in finitely many raw moments, so they are Lipschitz on the bounded moment range. Equations (3.4)–(3.5) and Theorem 4.1 now give

$$\|x - y\|_\infty \leq CH^{1/(m+1)}$$

in the unknown case, and the analogous bound with exponent $1/m$ in the known case. Taking square roots proves the scale estimates. Finally,

$$|v - w| \leq |\kappa_2(P_{a,v}) - \kappa_2(P_{b,w})| + \frac{m}{3} \|x - y\|_\infty$$

proves (6.3), absorbing the term CH since $H \leq \sqrt{2}$. Optimality follows from the pairs and asymptotics in Sections 7 and 9. $\square$

The inverse modulus alone does not prove a minimax sampling lower bound. One must construct pairs whose *likelihoods*, not merely a chosen finite list of moments, are close. That step is the subject of the next section.

7 From exact moment cancellation to likelihood asymptotics

Write $\phi_v(x) = (2\pi v)^{-1/2} \exp(-x^2/(2v))$. Fix a positive base variance v , let $f_0(x) = \mathbb{E}\phi_v(x - Z)$, and define

$$J_k(f_0) = \int_{\mathbb{R}} \frac{(f_0^{(k)}(x))^2}{f_0(x)} dx. \quad (7.1)$$

These quantities are finite despite the unbounded observation space.

Lemma 7.1 (Weighted derivative information). *For every integer $k \geq 1$,*

$$0 < J_k(f_0) \leq k!v^{-k}. \quad (7.2)$$

For $Z = 0$ equality holds.

Proof. Differentiating the Gaussian convolution is justified by its integrable Gaussian bounds. Conditional Cauchy–Schwarz gives pointwise

$$\frac{(\mathbb{E}\phi_v^{(k)}(x - Z))^2}{\mathbb{E}\phi_v(x - Z)} \leq \mathbb{E} \frac{(\phi_v^{(k)}(x - Z))^2}{\phi_v(x - Z)}.$$

Integrating and using Tonelli bounds $J_k(f_0)$ by the corresponding Gaussian integral. The derivative identity

$$\phi_v^{(k)}(x) = (-1)^k v^{-k/2} \text{He}_k(x/\sqrt{v})\phi_v(x)$$

and the Gaussian orthogonality relation $\mathbb{E} \text{He}_k(G)^2 = k!$ evaluate it as $k!v^{-k}$. Equality for $Z = 0$ is immediate. If $J_k(f_0) = 0$, its continuous numerator would vanish identically, so f_0 would be a polynomial of degree at most $k - 1$. No nonzero polynomial on $\mathbb{R}$ is an integrable probability density, a contradiction. $\square$

For any fixed nonnegative vector $u = (u_1, \dots, u_m)$ put

$$V_u = \sum_i \sqrt{u_i} U_i, \quad \beta_u = \frac{1}{3} p_1(u). \quad (7.3)$$

Two families will be used. The fixed-variance family is

$$f_{t,u}^{\text{fix}} = \text{density}(Z + tV_u + \sqrt{v}G). \quad (7.4)$$

The compensated family, defined for sufficiently small real t , is

$$f_{t,u}^{\text{comp}} = \text{density}(Z + tV_u + \sqrt{v - t^2\beta_u} G). \quad (7.5)$$

The total variance contributed by the added variables in the compensated family is exactly v for every t .

Theorem 7.2 (Exact leading likelihood term). *Consider either of the following cases:*

- (a) *The fixed family, with $q \geq 1$, and vectors u, w satisfying $p_k(u) = p_k(w)$ for $1 \leq k < q$.*
- (b) *The compensated family, with $q \geq 2$, and vectors u, w satisfying $p_k(u) = p_k(w)$ for $2 \leq k < q$.*

Assume $d = c_q(p_q(u) - p_q(w)) \neq 0$, and set $A = d/(2q)!$. Write $f_{t,u}, f_{t,w}$ for the chosen family. As $t \rightarrow 0$,

$$f_{t,u} - f_{t,w} = At^{2q}f_0^{(2q)} + O_{\mathcal{H}}(t^{2q+2}), \quad \mathcal{H} = L^2(\mathbb{R}, \phi_{3v/4}(x)^{-1} dx). \quad (7.6)$$

Moreover,

$$\chi^2(f_{t,u}, f_{t,w}) = A^2 J_{2q}(f_0) t^{4q} + o(t^{4q}), \quad (7.7)$$

$$H^2(f_{t,u}, f_{t,w}) = \frac{A^2}{4} J_{2q}(f_0) t^{4q} + o(t^{4q}). \quad (7.8)$$

In particular, both leading constants are strictly positive.

Proof. We give the tail and denominator details because they are essential for a statistical lower bound.

Step 1: a common positive lower envelope. Choose $|t| \leq t_0$ so that all Gaussian variances in the two families belong to $[7v/8, v]$. The shifts $W = Z + tV_u$ and $Z + tV_w$ are uniformly bounded by some M . Put $s = 3v/4$. For $b \in [7v/8, v]$ and $|W| \leq M$, minimization of a quadratic in x gives

$$\frac{\phi_b(x - W)}{\phi_s(x)} \geq \sqrt{\frac{s}{b}} \exp\left(-\frac{W^2}{2(b-s)}\right) \geq c > 0. \quad (7.9)$$

Averaging preserves the inequality. Thus each density in the two families, and f_0 , is bounded below by $c\phi_s$ uniformly in t .

Step 2: Taylor expansion in a weighted Hilbert space. Every fixed-order derivative in t of $\phi_{v-t^2\beta_u}(x - Z - tV_u)$, or of its fixed-variance counterpart, is a finite sum of Gaussian terms times polynomials in x, Z, V_u and bounded inverse powers of the variance. Consequently, for each derivative order it is bounded uniformly by

$$C(1 + |x|)^N \exp\left(-\frac{x^2}{2v} + C|x|\right). \quad (7.10)$$

After squaring and dividing by ϕ_s , the quadratic part of the exponent is $-x^2/(3v)$. This is integrable, including any polynomial and linear exponential factor. Differentiation under the expectation and differentiation in $\mathcal{H}$ are therefore justified to any fixed order. Symmetry of V_u makes the density an even function of t . Taylor's theorem in $\mathcal{H}$ gives an expansion with an $O_{\mathcal{H}}(t^{2q+2})$ remainder through order $2q$.

Step 3: identify the first unequal coefficient. For each fixed Fourier frequency ξ , the logarithm of the uniform part has its convergent germ

$$\log \mathbb{E} e^{i\xi t V_u} = \sum_{k \geq 1} \frac{c_k p_k(u)}{(2k)!} (i\xi t)^{2k}.$$

In the compensated family the Gaussian adjustment cancels the $k = 1$ term exactly. The moment equalities in (a) or (b) then show that the first unequal coefficient of the two characteristic functions is

$$At^{2q}(i\xi)^{2q} \widehat{f_0}(\xi).$$

Here the characteristic-function convention is $\widehat{f}(\xi) = \int e^{i\xi x} f(x) dx$; the Fourier transform of an even $2q$ th derivative has the same multiplier $(i\xi)^{2q}$. No logarithm of $\widehat{f}_0$ is needed, so its possible zeros cause no problem. Coefficients in the Hilbert-space expansion are integrable by Cauchy–Schwarz against ϕ_s , and Fourier uniqueness identifies their difference as $Af_0^{(2q)}$. This proves (7.6).

Step 4: pass to the likelihood metrics. Let $D_t = (f_{t,u} - f_{t,w})/t^{2q}$ and $D = Af_0^{(2q)}$. We know $D_t \rightarrow D$ in $\mathcal{H}$. The weights $\phi_s/f_{t,w}$ are uniformly bounded by (7.9) and converge pointwise to ϕ_s/f_0 . Weighted Cauchy–Schwarz first replaces D_t by D in the integral, and dominated convergence then gives

$$\int \frac{D_t^2}{f_{t,w}} dx \rightarrow \int \frac{D^2}{f_0} dx.$$

This is (7.7). Finally, the identity

$$(\sqrt{f} - \sqrt{g})^2 = \frac{(f - g)^2}{(\sqrt{f} + \sqrt{g})^2}$$

has denominator converging pointwise to $4f_0$ and bounded below by $4c\phi_s$. The identical argument gives (7.8). Positivity of the limits follows from Lemma 7.1. $\square$

This proof explains why merely matching finitely many moments is not enough: the required control is in a space weighted by a reciprocal density. The positive Gaussian component supplies a common tail envelope and prevents rare tail events from defeating the lower-bound construction.

8 A feasible moment estimator and upper bounds

This section constructs a measurable estimator and proves the upper half of Theorem 2.1. The procedure is deliberately elementary and globally specified. It is not claimed to be computationally optimal.

8.1 Uniform moment control and cumulant estimation

For every integer $j \geq 1$, all parameters in the unknown model satisfy

$$\mathbb{E}|X_{a,v}|^j \leq B_j := 2^{j-1} \left((L+m)^j + \bar{v}^{j/2} \mathbb{E}|G|^j \right). \quad (8.1)$$

For known variance use v_0 instead of $\bar{v}$. Given observations $X_1, \dots, X_n$, set $\widehat{\mu}_0 = 1$ and

$$\widehat{\mu}_j = \text{clip}_{[-B_j, B_j]} \left(\frac{1}{n} \sum_{\ell=1}^n X_\ell^j \right). \quad (8.2)$$

Because the true raw moment lies in the clipping interval,

$$\mathbb{E}|\widehat{\mu}_j - \mu_j|^2 \leq \frac{B_{2j}}{n}. \quad (8.3)$$

Cumulants are computed recursively from these clipped moments:

$$\widehat{\kappa}_j = \widehat{\mu}_j - \sum_{\ell=1}^{j-1} \binom{j-1}{\ell} \widehat{\kappa}_\ell \widehat{\mu}_{j-\ell}. \quad (8.4)$$

The exact identity follows by differentiating the relation between a moment generating function and its logarithm, then comparing coefficients. Its estimated version is a polynomial in a bounded moment vector. Every fixed polynomial is Lipschitz on this compact box, so for fixed K ,

$$\mathbb{E} \max_{1 \leq j \leq K} |\widehat{\kappa}_j - \kappa_j(X_{a,v})|^2 \leq \frac{C_K}{n}. \quad (8.5)$$

The estimators need not be unbiased; the bound, not unbiasedness, is what the inversion argument requires.

For unknown variance use $K = 2m + 2$ and define

$$\hat{p}_k = \frac{\hat{\kappa}_{2k} - \kappa_{2k}(Z)}{c_k}, \quad 2 \leq k \leq m + 1. \quad (8.6)$$

For known variance use $K = 2m$, the same formula for $2 \leq k \leq m$, and

$$\hat{p}_1 = 3(\hat{\kappa}_2 - \kappa_2(Z) - v_0). \quad (8.7)$$

With $r = 1$ in the unknown case and $r = 0$ in the known case, write

$$\Delta = \max_{r+1 \leq k \leq r+m} |\hat{p}_k - p_k(x)|. \quad (8.8)$$

Equations (8.5) and $c_k \neq 0$ imply

$$\mathbb{E}\Delta^2 \leq C/n, \quad \mathbb{E}\Delta \leq Cn^{-1/2}. \quad (8.9)$$

8.2 Globally feasible fitting instead of unstable root clipping

A noisy moment vector need not correspond to nonnegative real nodes. We therefore fit within the admissible set rather than form a polynomial and silently discard its complex or negative roots. Let

$$\mathcal{G}_n = \mathcal{X}_m \cap \{0, 1/n, \dots, 1\}^m. \quad (8.10)$$

Choose the lexicographically first minimizer

$$\hat{x} \in \arg \min_{z \in \mathcal{G}_n} \max_{r+1 \leq k \leq r+m} |p_k(z) - \hat{p}_k|, \quad \hat{a}_i = \sqrt{\hat{x}_i}. \quad (8.11)$$

There are exactly $\binom{n+m}{m}$ grid points, so the rule exists and is measurable. It is an exhaustive finite algorithm when the required constants and objective comparisons are provided to the desired accuracy. Its grid size is a statistical existence device, not a recommendation for large- m numerical implementation.

Coordinatewise downward rounding of the true x to this grid preserves order and changes each coordinate by at most $1/n$. Since $z \mapsto z^k$ is k -Lipschitz on $[0, 1]$, the rounded vector has objective at most $\Delta + m(m+r)/n$. The minimizer therefore satisfies

$$E_r(\hat{x}, x) \leq 2\Delta + \frac{m(m+r)}{n}. \quad (8.12)$$

Theorem 4.1 and Corollary 4.2 yield

$$\|\hat{x} - x\|_\infty \leq C(\Delta + n^{-1})^{1/(m+r)}, \quad (8.13)$$

$$\|\hat{a} - a\|_\infty \leq C(\Delta + n^{-1})^{1/(2m+2r)}. \quad (8.14)$$

Jensen's inequality and (8.9) prove

$$\sup \mathbb{E} \|\hat{a} - a\|_\infty \leq Cn^{-1/(4m+4r)}. \quad (8.15)$$

Taking $r = 0, 1$ gives the two scale upper bounds.

In the unknown case define the variance estimator

$$\hat{v} = \text{clip}_{[v, \bar{v}]} \left(\hat{\kappa}_2 - \kappa_2(Z) - \frac{1}{3}p_1(\hat{x}) \right). \quad (8.16)$$

Clipping cannot increase its distance from the true v . Thus

$$\mathbb{E}|\hat{v} - v| \leq \mathbb{E}|\hat{\kappa}_2 - \kappa_2(X)| + \frac{m}{3} \mathbb{E} \|\hat{x} - x\|_\infty \leq Cn^{-1/(2m+2)}, \quad (8.17)$$

where the faster $n^{-1/2}$ term is absorbed. This completes all upper bounds in Theorem 2.1.

8.3 Approximate optimization and confidence bounds

The estimator remains valid if the achieved objective is within η_n of the global grid minimum. Equation (8.12) then acquires an additional η_n . In particular, a certified objective gap $\eta_n = O(n^{-1/2})$ preserves the rates. An arbitrary local optimum, without a bound on its gap, has no such guarantee.

For $0 < \delta < 1$, Markov's inequality applied to Δ^2 gives, with probability at least $1 - \delta$,

$$\|\hat{a} - a\|_\infty \leq C \left((n\delta)^{-1/2} + n^{-1} \right)^{1/(2m+2r)}. \quad (8.18)$$

These elementary confidence bounds are conservative. Sharper concentration or adaptive confidence procedures are separate questions; neither is needed for the minimax mean-risk result.

9 Matching statistical lower bounds

We first record the testing reduction with the metric conventions used here. If two parameters are separated by D in a metric loss, the test that chooses the parameter nearer an estimator has an error only if that estimator's loss is at least $D/2$. Consequently, for every estimator,

$$\max_{i=0,1} \mathbb{E}_i[\text{loss}] \geq \frac{D}{4} (1 - d_{\text{TV}}(P_0^{\otimes n}, P_1^{\otimes n})). \quad (9.1)$$

This follows because the minimum sum of the two simple testing errors is $1 - d_{\text{TV}}$. Hellinger affinity tensorizes, so

$$H^2(P^{\otimes n}, Q^{\otimes n}) = 2 \left[1 - \left(1 - \frac{H^2(P, Q)}{2} \right)^n \right] \leq nH^2(P, Q). \quad (9.2)$$

Together with $d_{\text{TV}} \leq H$, this controls the bracket in (9.1).

9.1 Unknown Gaussian variance

Choose $v_* \in (\underline{v}, \bar{v})$ and the fixed positive distinct vectors u, w from Proposition 5.2, with a sufficiently small nonzero flow parameter. They satisfy (5.11). For small $t > 0$ define two model parameters by

$$a_i(t) = t\sqrt{u_i}, \quad v_u(t) = v_* - \frac{t^2}{3}p_1(u), \quad (9.3)$$

$$b_i(t) = t\sqrt{w_i}, \quad v_w(t) = v_* - \frac{t^2}{3}p_1(w). \quad (9.4)$$

They belong to the parameter set for every sufficiently small t , including the variance interval. Their scale and variance separations are

$$\|a(t) - b(t)\|_\infty = D_a t, \quad |v_u(t) - v_w(t)| = D_v t^2, \quad (9.5)$$

where $D_a, D_v > 0$ are fixed. Theorem 7.2, with $q = m + 1$, gives

$$H^2(P_{a(t), v_u(t)}, P_{b(t), v_w(t)}) = K t^{4m+4} + o(t^{4m+4}), \quad K > 0. \quad (9.6)$$

Set $t = \gamma n^{-1/(4m+4)}$ with fixed $\gamma > 0$ small enough. Equations (9.2) and (9.6) ensure product total variation at most $1/2$ for all sufficiently large n . Applying (9.1) separately to the two separations in (9.5) proves

$$\mathcal{R}_n^a \geq c n^{-1/(4m+4)}, \quad \mathcal{R}_n^v \geq c n^{-1/(2m+2)}. \quad (9.7)$$

The same pair proves both lower bounds; the variance difficulty is not an artifact of the plug-in estimator.

9.2 Known Gaussian variance

Use the construction of Lemma 5.1 with $r = 0$ to choose positive vectors u, w for which $p_1, \dots, p_{m-1}$ agree but p_m differs. For $m = 1$ there are no matching constraints, and any two distinct positive nodes suffice. Set $a = t\sqrt{u}$, $b = t\sqrt{w}$, and keep the Gaussian variance fixed at v_0 . Theorem 7.2, now with $q = m$ in the fixed family, gives $H^2 \asymp t^{4m}$. Choosing $t = \gamma n^{-1/(4m)}$ in the same two-point argument proves

$$\mathcal{R}_n^{a,\text{known}} \geq cn^{-1/(4m)}. \quad (9.8)$$

This completes Theorem 2.1.

9.3 Optimality of the deterministic moduli

The same pairs also complete Theorem 6.2. In the unknown case $H \asymp t^{2m+2}$ while scale distance is $D_a t$ and variance distance is $D_v t^2$. An exponent larger than $1/(2m+2)$ or $1/(m+1)$, respectively, would contradict these asymptotics. In the known case $H \asymp t^{2m}$ and scale distance is proportional to t , excluding a larger exponent there.

9.4 Why these are global, not everywhere-local, rates

The hard parameters approach the boundary where all uniform factors vanish. They do not say that well-separated positive factors are always recovered slowly. In fact, for the unknown-variance moment map $F(x) = (p_2(x), \dots, p_{m+1}(x))$,

$$\det DF(x) = (m+1)! \left(\prod_{i=1}^m x_i \right) \prod_{1 \leq i < j \leq m} (x_j - x_i). \quad (9.9)$$

To check this, factor k from the row for p_k and x_i from column i ; the remaining determinant is Vandermonde. At a strictly positive, distinct configuration this determinant is nonzero. The inverse function theorem, together with the moment bounds of Section 8, yields local $n^{-1/2}$ estimation on a sufficiently small regular neighborhood, including for v and for the half-lengths. The slow global exponents reflect loss of rank near vanishing or colliding configurations, not a uniform penalty at every point.

10 Detection, rank, and the distinction from recovery

We prove Theorem 2.2. For unknown variance the exact fourth-cumulant contrast is

$$T = \kappa_4(X_{a,v}) - \kappa_4(Z) = -\frac{2}{15} \sum_i a_i^4. \quad (10.1)$$

Let $\hat{T} = \hat{\kappa}_4 - \kappa_4(Z)$, using the clipped-moment procedure. By (8.5), $\mathbb{E}|\hat{T} - T|^2 \leq C/n$ uniformly. Reject $a = 0$ when

$$\hat{T} < -h^4/15. \quad (10.2)$$

Under the null $T = 0$; under any alternative with $\|a\|_\infty \geq h$, $T \leq -2h^4/15$. Either kind of error requires $|\hat{T} - T| \geq h^4/15$. Hence the sum of maximal errors is at most $C/(nh^8)$.

For the converse, fix $v_* \in (\underline{v}, \bar{v})$ and compare

$$(a, v) = (0, v_*) \quad \text{with} \quad (a, v) = ((0, \dots, 0, h), v_* - h^2/3). \quad (10.3)$$

For small h both are admissible. Apply Theorem 7.2 with $q = 2$, one node equal to 1 versus all nodes zero, and $t = h$. There are no higher matching constraints before p_2 ; compensation matches variance.

The one-observation squared Hellinger distance is $O(h^8)$. If $nh_n^8 \rightarrow 0$, product total variation tends to zero by (9.2), so every test has the sum of its two errors at least $1 - o(1)$. Composite maximal risks are at least these two simple risks.

When variance is known, use

$$T_2 = \kappa_2(X_{a,v_0}) - \kappa_2(Z) - v_0 = \frac{1}{3} \sum_i a_i^2 \quad (10.4)$$

and reject for $\hat{T}_2 > h^2/6$. This gives $C/(nh^4)$ by the same mean square argument. The fixed-variance one-factor pair has $H^2 = O(h^4)$ by Theorem 7.2 with $q = 1$, proving the lower bound.

At a fixed large multiple of the critical scale the upper bounds can be made arbitrarily small by increasing that multiple. Consistency, meaning errors tending to zero with n , follows for diverging multiples. We do not confuse these two statements.

Corollary 10.1 (No uniform exact factor-count estimation without separation). *Let the factor count be $\#\{i : a_i > 0\}$. Over either full parameter set of Section 2, no sequence of estimators of this count has probability of error tending to zero uniformly over all parameters.*

Proof. Choose h_n below the respective detection scale in (10.3), or in its known-variance counterpart. The two parameters have different counts, but their product laws have total variation tending to zero. An exact count estimator would distinguish them, contradicting the testing lower bound. $\square$

This does not rule out pointwise consistency when every nonzero factor has a fixed positive half-length. It rules out *uniform* recovery of the count without a lower bound on the smallest positive half-length.

11 Explicit two-factor and one-factor examples

11.1 A two-factor pair with a sixth-order first difference

Take squared-scale templates

$$u = (0, 5), \quad w = (3, 4). \quad (11.1)$$

The templates themselves exceed 1, but the actual squared scales are $t^2 u, t^2 w$, which lie in $[0, 1]^2$ for sufficiently small t . They satisfy

$$p_1(u) = 5, \quad p_1(w) = 7, \quad p_2(u) = p_2(w) = 25, \quad p_3(u) - p_3(w) = 34. \quad (11.2)$$

Choose Gaussian variances $v_* - 5t^2/3$ and $v_* - 7t^2/3$. The two uniform half-length vectors are $(0, \sqrt{5}t)$ and $(\sqrt{3}t, 2t)$, separated by $\sqrt{3}t$, while the variances differ by $2t^2/3$. The first unequal added cumulant is the sixth:

$$\Delta\kappa_6 = \frac{16}{63} 34t^6 = \frac{544}{63} t^6. \quad (11.3)$$

Theorem 7.2 therefore gives

$$f_{t,u} - f_{t,w} = \frac{34}{2835} t^6 f_0^{(6)} + O_{\mathcal{H}}(t^8). \quad (11.4)$$

For $Z = 0$ and $v_* = 1$, the corresponding chi-square expansion is

$$\chi^2(f_{t,u}, f_{t,w}) = \frac{18496}{178605} t^{12} + o(t^{12}). \quad (11.5)$$

Indeed, $J_6(\phi_1) = 6! = 720$, and $720(34/2835)^2 = 18496/178605$. This gives the unknown-variance two-factor scale $t \asymp n^{-1/12}$ and variance separation of order $n^{-1/6}$ without requiring a numerically solved polynomial.

For a known-variance comparison, use $u = (1, 4)$ and $w = (2, 3)$. Their first power sums agree and their second power sums differ by 4. The leading density difference is

$$f_{t,u}^{\text{fix}} - f_{t,w}^{\text{fix}} = -\frac{1}{45}t^4 f_0^{(4)} + O_{\mathcal{H}}(t^6), \quad (11.6)$$

which gives the faster two-factor scale $n^{-1/8}$.

11.2 One small factor and an unknown Gaussian variance

Compare $\sqrt{v_*}G$ with $tU + \sqrt{v_* - t^2/3}G$. Variance is exactly matched, and the fourth cumulant is the first non-Gaussian signal:

$$f_t - \phi_{v_*} = -\frac{t^4}{180}\phi_{v_*}^{(4)} + O_{\mathcal{H}}(t^6), \quad \chi^2(f_t, \phi_{v_*}) = \frac{4!}{180^2 v_*^4} t^8 + o(t^8). \quad (11.7)$$

This is the simplest instance of the $n^{-1/8}$ threshold. If the variance cannot adjust, the leading density difference is instead $t^2 \phi_{v_*}''/6$, giving an $n^{-1/4}$ threshold. The nuisance parameter removes a genuine leading observable term.

11.3 A Fourier diagnostic with a known exact limit

For the pair (11.1), $Z = 0$, $v_* = 1$, and frequency 1, put

$$\ell_u(t) = -\frac{1 - 5t^2/3}{2} + \log \text{sinc}(\sqrt{5}t), \quad (11.8)$$

$$\ell_w(t) = -\frac{1 - 7t^2/3}{2} + \log \text{sinc}(\sqrt{3}t) + \log \text{sinc}(2t). \quad (11.9)$$

For small positive t the sinc factors are positive, so these real logarithms are unambiguous. The proved coefficient identity gives

$$\frac{\ell_u(t) - \ell_w(t)}{t^6} \longrightarrow -\frac{34}{2835}. \quad (11.10)$$

The included program evaluates this diagnostic at 80-digit working precision:

t	$(\ell_u(t) - \ell_w(t))/t^6$
0.2	-0.0123041895267270
0.1	-0.0120695349593127
0.05	-0.0120120178007690
0.025	-0.0119977087830769
0.0125	-0.0119941358994491

The calculation is a diagnostic, not an interval-certified proof of the asymptotic statement or of the likelihood integrals.

12 Verification record and a formalization route

12.1 What the accompanying program checks

The package contains `verify_results.py`, requiring `SymPy` and `mpmath`, with random seed 20260929. It produces a machine-readable JSON record and a plain-text summary. The completed checks are:

1. 7000 exact rational instances of (4.2), for $1 \leq m \leq 7$, $0 \leq r \leq 3$, including zero entries and repeated nodes. Each comparison uses integer or rational arithmetic, not floating-point Hölder powers.
2. 40 exact Lagrange/Vandermonde tangent checks for $1 \leq m \leq 8$ and $0 \leq r \leq 4$.
3. Eight exact coefficient-flow checks, one for each $1 \leq m \leq 8$. For every case, a nonzero rational shift is accompanied by m disjoint positive rational intervals on each of which the resulting degree- m polynomial changes sign. This certifies all its roots to be real, positive, and distinct. Power-sum conservation and the derivative of p_{m+1} are verified symbolically.
4. Exact evaluation of the two-factor leading constants in (11.4)–(11.5), followed by the high-precision Fourier diagnostic above.

Every listed exact check passed. These finite checks support the algebra and examples but do not replace the all-parameter proofs. The program does not claim a simulated minimax theorem, an empirical risk study, an interval bound on an integral, or a proof-assistant verification.

12.2 A useful division of the formalization work

The argument separates into three fairly independent layers.

Finite algebra and order. Formalize ordered nonnegative vectors, power sums, counting functions, their sign blocks, and Theorem 4.1. The bottleneck is not symbolic manipulation but the finite combinatorial bound on sign changes, including ties. Lagrange interpolation and the elementary-symmetric flow can then be proved as exact polynomial identities. The program’s rational root intervals are natural certificate data for this layer, provided a proved checker supplies their semantics.

Analysis of Gaussian convolutions. Formalize the common lower envelope, Gaussian derivative/Hermite bounds, parameter differentiation in a weighted L^2 space, and the passage from that convergence to chi-square and Hellinger limits. This is the essential analytic bridge; an unproved assertion that moment matching implies Hellinger closeness must not substitute for it.

Statistical decisions. Formalize measurable finite-grid minimization, uniform moment bounds, clipping, the two-point testing inequality, and Hellinger tensorization. Once these are available, the exponents follow by short substitutions. Theorems about the known and unknown variance experiments should retain different parameter types or explicit hypotheses, preventing an accidental removal of the positive-smoothing assumption.

This is a roadmap, not delivered Lean code. The repository’s existing formal Fabius tools motivate the background, but no assertion in this article is promoted to a formal theorem merely because an adjacent repository module has been kernel-checked.

13 Further research questions and success criteria

Research question 1 (Remove Gaussian smoothing without losing the lower bound). For the unsmoothed model $Z_{\text{up}} + \sum_i a_i U_i$, determine the sharp sampling minimax rates at vanishing factors. The key issue is whether a suitable replacement for the weighted Taylor argument holds near the flat, moving support boundary. A first concrete target is finiteness, or a precise failure mode, of $\int (g^{(k)})^2 / g$ for the up density g , together with control of the support change under convolution. A total-variation inverse modulus alone does not settle this problem.

Research question 2 (Classify all local strata with unknown variance). At a base configuration with positive squared-scale multiplicities $m_1, \dots, m_s$ and m_0 zero slots, determine the exact local pairwise Hellinger inverse exponent. A candidate denominator for the half-length exponent is

$$M = \max(\{m_1, \dots, m_s\} \cup \{2(m_0 + 1) : m_0 > 0\}). \quad (13.1)$$

The proposed exponent is $1/M$ for the Hellinger modulus and $1/(2M)$ for the corresponding sampling rate. This is a conjectural extension, not a result proved here. It agrees with the fully vanishing and regular cases proved above, but mixed clusters require a separate local argument and matching likelihood constructions.

Editorial note (ProveIt, 2026-09-29). The later repository article `inverse-and-sampling/Local_Minimax_Geometry_Uniform_Factors/` (filed 2026-09-29, unreviewed) proves the candidate denominator (13.1) at the level of minimax sampling rates, $n^{-1/(2M)}$, for Gaussian variance unknown in a compact interval bounded away from zero (its Corollary `cor: joint`; its r is m_0 here); it does not cite this problem. The pairwise Hellinger-modulus form is not stated there. The same article constructs one grid moment estimator attaining the local rates on every fixed stratum without being told the stratum, which bears on the next problem; that estimator is not efficient, and honest adaptive confidence sets remain open.

Research question 3 (Adapt to collision and vanishing patterns). Construct a single practical estimator attaining the regular $n^{-1/2}$ rate away from singularities and the correct slower rates on every singular stratum without being told the stratum. Determine which honest confidence sets can adapt to these rates. Separate impossibility due to testing between strata from failures of a particular algorithm.

Research question 4 (Efficient globally certified shifted-moment fitting). Replace the $\binom{n+m}{m}$ grid search by an algorithm with explicit complexity in m and the requested objective tolerance. One route is to use hyperbolic polynomial coefficients and the special flow of Proposition 5.2; another is constrained optimization with a certificate of a global objective gap. A useful endpoint is a solver whose certificate guarantees the $O(n^{-1/2})$ gap sufficient for the statistical rates, including multiple and zero roots.

Research question 5 (Growing capacity and the infinite-factor transition). Allow $m = m_n \rightarrow \infty$ or a summable infinite half-length spectrum. Track constants in the shifted-moment, cumulant-estimation, and likelihood bounds instead of hiding them in fixed- m notation. Determine where the polynomial rates proved here cease to be informative and connect that regime to the repository's infinite-factor instability questions. The fixed- m theorem cannot simply be read with a growing m substituted into its exponent.

Editorial note (ProveIt, 2026-09-29). For the Gaussian variance alone, the later repository article `inverse-and-sampling/Gaussian_Dust_Christoffel_Recovery_Uniform_Factors/` (filed 2026-09-29, unreviewed) treats square-summable infinite half-length spectra: with total latent variance at most V , the exact minimax risk for the variance is $V/2$ in absolute error at every sample size, and uniform consistency holds exactly on classes whose squared-scale tails vanish uniformly. Growing capacity $m = m_n$ and the recovery of the half-lengths are not treated there.

Research question 6 (Learn the background from calibration samples). Suppose the background law is not known exactly but a second independent sample from Z is available. Quantify the joint dependence on the observation sample size and calibration sample size, particularly when high cumulants of Z must be estimated. A second variant is an explicitly bounded deterministic error in each background cumulant. Determine when background uncertainty becomes the dominant obstruction.

Research question 7 (Multivariate uniform directions and covariance confounding). Replace each $a_i U_i$ by $U_i b_i$ with unknown vectors $b_i \in \mathbb{R}^d$, and let the Gaussian covariance matrix be unknown. Higher even cumulants are then sums of tensors $b_i^{\otimes 2k}$. Establish identifiability up to signs and permutations, and sharp rates near collinear, colliding, or vanishing vectors. The one-dimensional sign-count proof no longer applies directly; it isolates what a genuine multivariate replacement must accomplish.

Research question 8 (Optimal constants and different moment windows). Determine the best constant in (4.2), including its dependence on m, r . More structurally, classify which nonconsecutive power windows retain injectivity and what their optimal inverse exponents are. Then examine known unequal positive weights in the power sums. Equal unit weights were used in the excursion-count argument, so a weighted extension needs its own hypotheses rather than an automatic transfer.

14 Conclusion

Unknown Gaussian variance does more than add an extra parameter. It absorbs the second-cumulant signature of small uniform factors, so the recoverable power-sum window shifts from $p_1, \dots, p_m$ to $p_2, \dots, p_{m+1}$. The sharp shifted-moment inequality quantifies that loss algebraically; exact moment fibres and a weighted Gaussian likelihood expansion show that it persists at the level of statistical information. The resulting scale rates are $n^{-1/(4m)}$ with known variance and $n^{-1/(4m+4)}$ with unknown variance, with variance estimation at $n^{-1/(2m+2)}$. Detection uses only the first unconfounded non-Gaussian cumulant and has the different, capacity-free threshold $n^{-1/8}$.

These conclusions extend a specific ProveIt research direction without assuming the unreviewed preceding report as a proved analytic black box. They leave explicit boundaries: positive smoothing, fixed capacity, known bounded background, and conventional rather than machine-checked proofs. The most direct next advance is a sharp treatment of mixed local strata or of the unsmoothed up-law likelihood boundary.

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